



REVERIE

What should I watch next?

Deep Learning · IE University · Functional MVP

Amaly Attia · Em Echeverria · Lea Sarouphim Hochar · Stephan Pentchev · Cécile Tambey

More choice made watching harder, not easier

Tens of thousands of titles.

Ten minutes deciding.

Then you give up.



Decision fatigue is real

- More choice didn't help — it overwhelmed. Endless scrolling replaced watching, and the joy of picking a film turned into a chore.



For platforms, fatigue = churn

- When a viewer can't decide, they close the app. Repeated, that indecision becomes cancelled subscriptions — the metric streaming services fear most.

Your next film, in under a second



Import what you've seen, and Reverie ranks the whole catalog to your taste — instantly.



Import your history

- Bring your Letterboxd or Netflix history. No cold quiz, no manual rating — just upload and go.



Ranked to your taste

- The model scores every film in the catalog against what you actually love, then hands you a short, trustworthy list.



The Blend

- Movie night for two? The Blend finds picks that satisfy both viewers at once — one list, two tastes.

PROOF IT WORKS

1.00

precision on the high-confidence “you will love it” picks

Across held-out real viewers — when Reverie is sure, it has been right. Every film it flagged as a “love” was one.



Why better recommendations pay for themselves

~\$1B

per year

Netflix attributes this to its recommender, via reduced churn.

~€1.8M

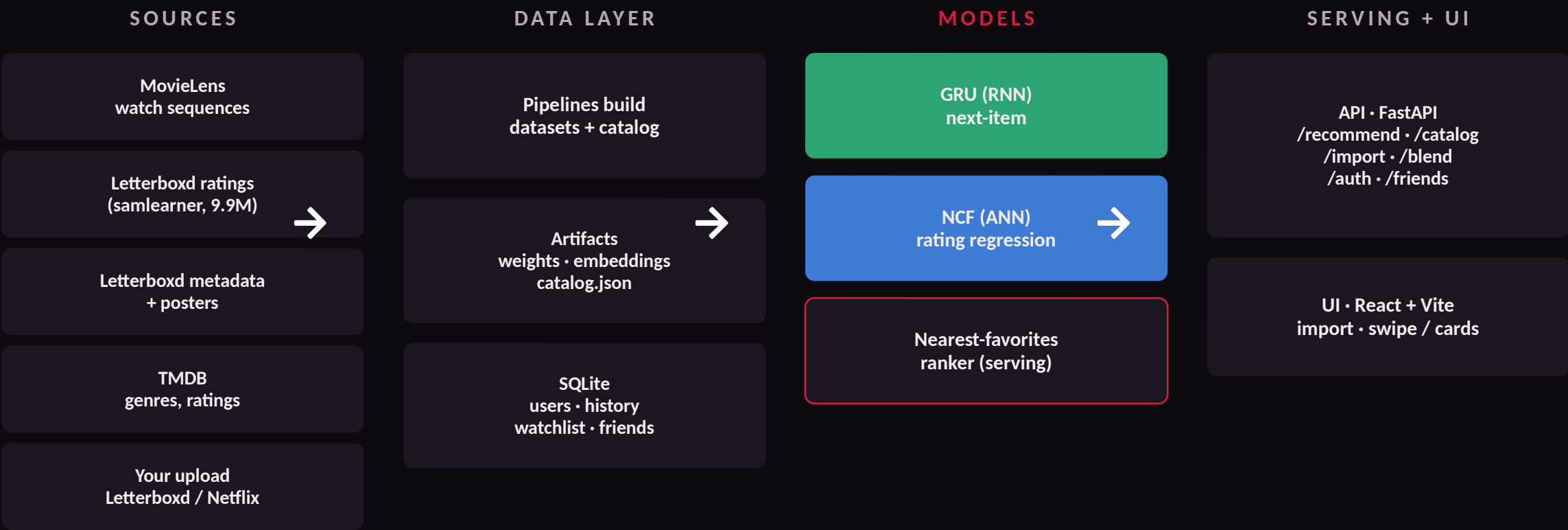
per year

Retained from a **1-point churn cut** on 1M subscribers at €15/mo. A tiny relevance gain pays for the whole system.

01 How we built it

The system, end to end

Sources → Data → Models → Product



Two predictive networks (an RNN and an ANN) feed a serving layer; a FastAPI backend and a React frontend make it a real product.

One product, two questions



Model 1 • GRU

“What comes next in your sequence?”

- Recurrent network (RNN) — order matters
- Sequential: reads your history in order
- Classification: softmax over the catalog
- Trained on MovieLens watch sequences



Model 2 • NCF

“How much will you like this film?”

- Feed-forward network (ANN) with embeddings
- Collaborative: learns from everyone
- Regression: a predicted rating 1–10
- Trained on 9.9M Letterboxd ratings

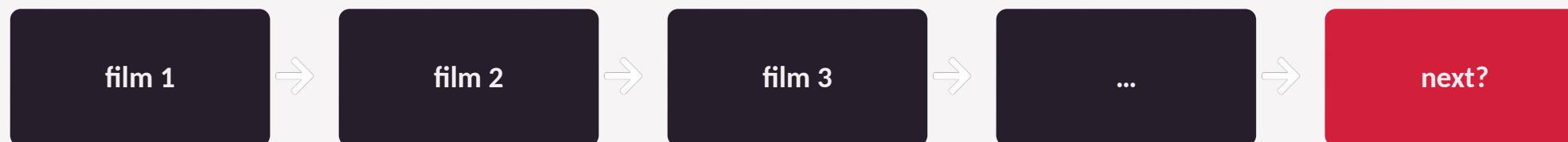
A cell that remembers



The memory loop: each film's features update a hidden state h_t that is passed to the next step. The hidden state carries everything the model has seen so far.

The head is a classification over the catalog, trained with cross-entropy loss. (Concepts from class: embeddings, the memory loop, softmax.)

Trained on every prefix, evaluated leave-one-out



All-prefix training

- From one history we learn every prefix → next-film pair, so one viewer becomes many training examples.
- Leave-one-out evaluation: the last film per user is held out and never seen in training.



Config chosen on validation

- L = 20 history window, embedding 32, 64 GRU units.
- GRU chosen over LSTM — same score, fewer parameters.
- Train-only statistics — no leakage into the held-out set.

Embeddings into hidden layers



The Dense + ReLU layers are the hidden layers that learn what makes you you. **The loss is MSE, reported as RMSE.**

Biases recover the baseline, embeddings learn the residual

$$\text{rating} = \text{global mean} + \text{user bias} + \text{movie bias} + \text{MLP}(\mathbf{u}, \mathbf{m}, \text{content})$$



Biases do the easy work

- The global mean plus per-user and per-movie biases recover the obvious baseline rating — cheaply and stably.



Embeddings learn the rest

- The neural network (MLP) learns the residual — the subtle, personal signal that makes your taste yours.

We rank by your nearest favorites



An honest design choice: a cold fold-in (building a fresh user vector from a few ratings) underperformed, so we don't ship it.



What we ship instead

- Rank each film by its similarity to your highest-rated films, in the learned movie-embedding space.
- Plus a genre anchor and a recency nudge to keep it fresh.



Why this is the right call

- The movie-embedding space is well-trained and reliable.
- A fresh user vector from only a few ratings is not — so we lean on what's trustworthy. A choice we reached by testing.

02 Evidence

Does it work?

We watched it overfit, then stopped



Early stop • epoch 16

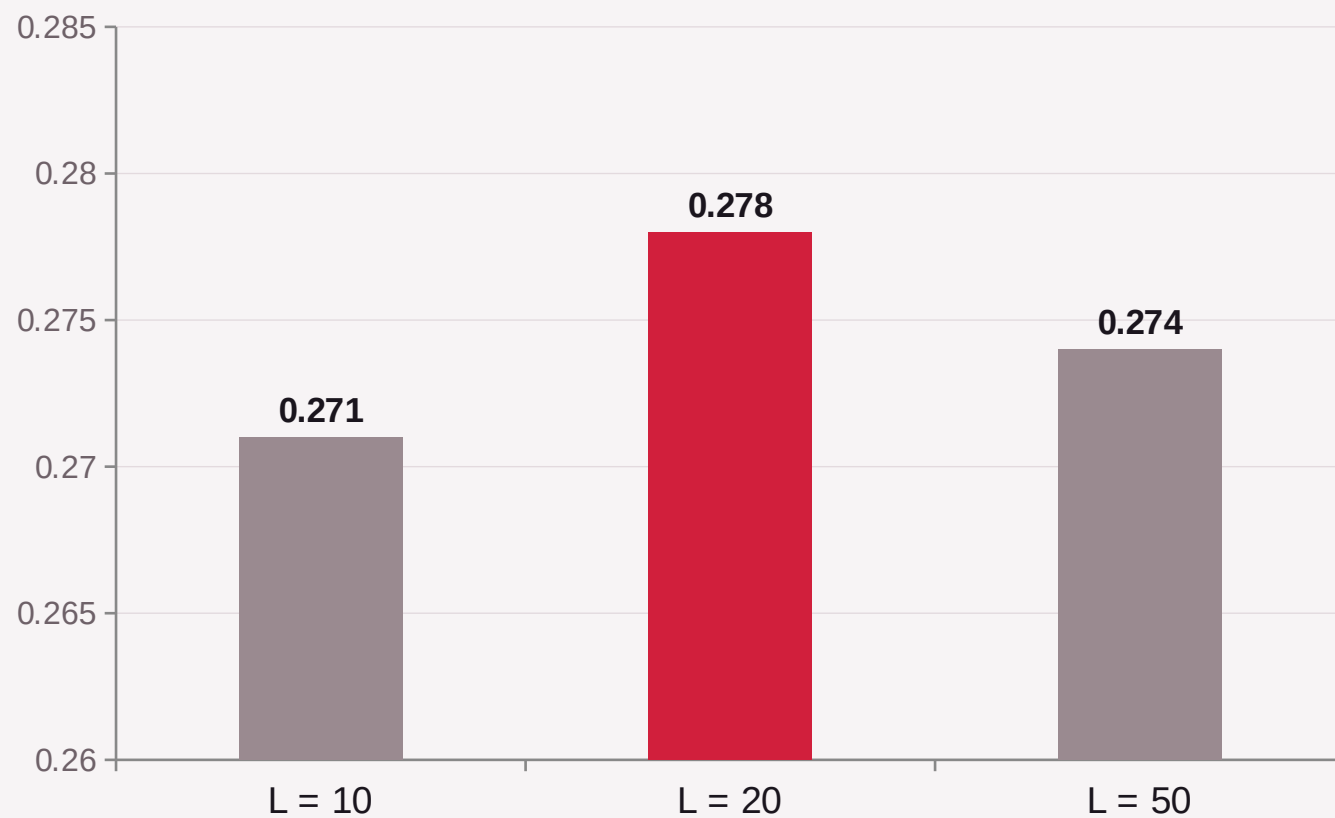
- Validation loss bottoms at epoch 16, then climbs. We stop there and restore the best weights.



The real run, not theory

- Training loss keeps falling while val turns up — the textbook overfitting signature, caught live.

History length: 20 wins



E3 • history length

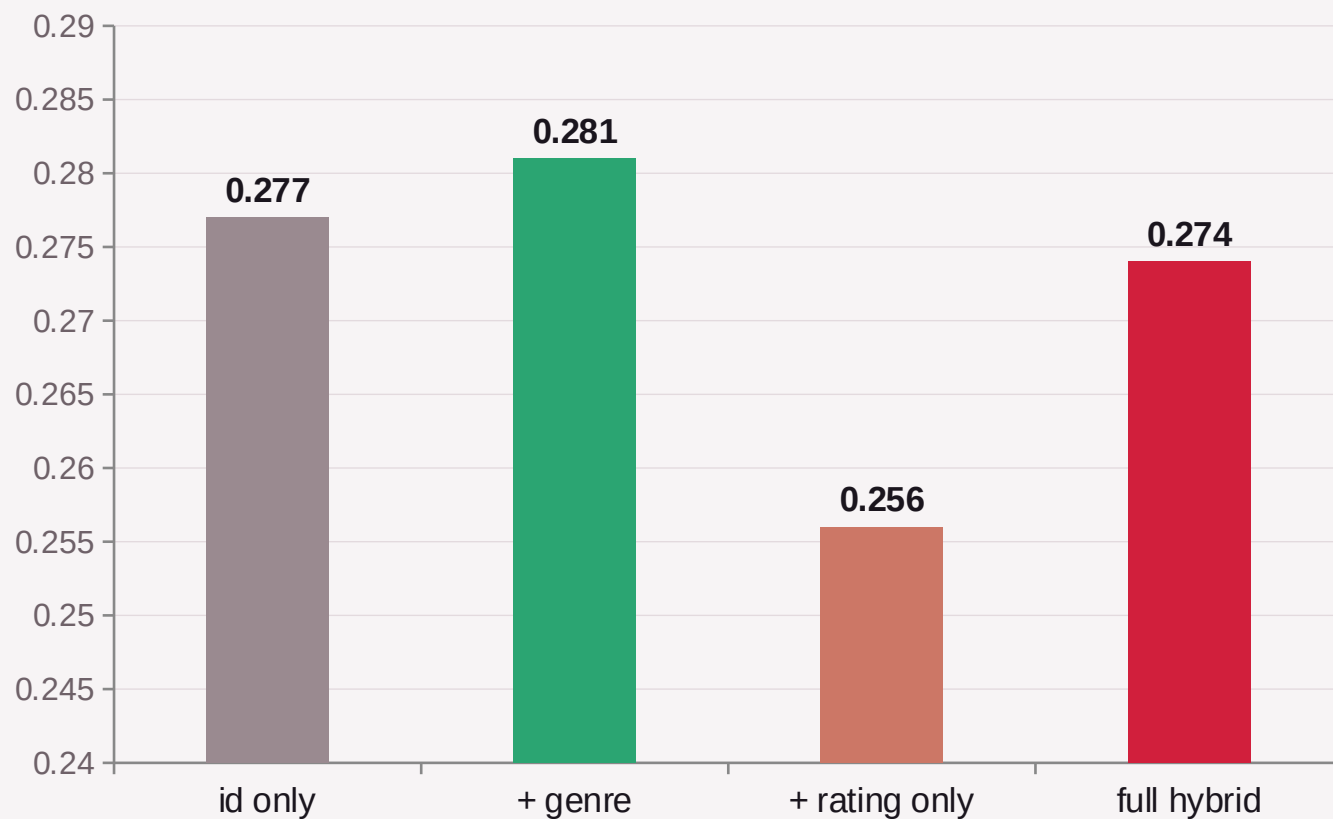
- 20 beats 10 and 50 on HR@10 — recent taste carries the signal.



E4 + E5 decisions

- Kept d32 / u64 (smaller train-val gap than bigger nets); GRU over LSTM (same score, fewer params).

Genre helps. Rating alone hurts.



Why ship the hybrid?

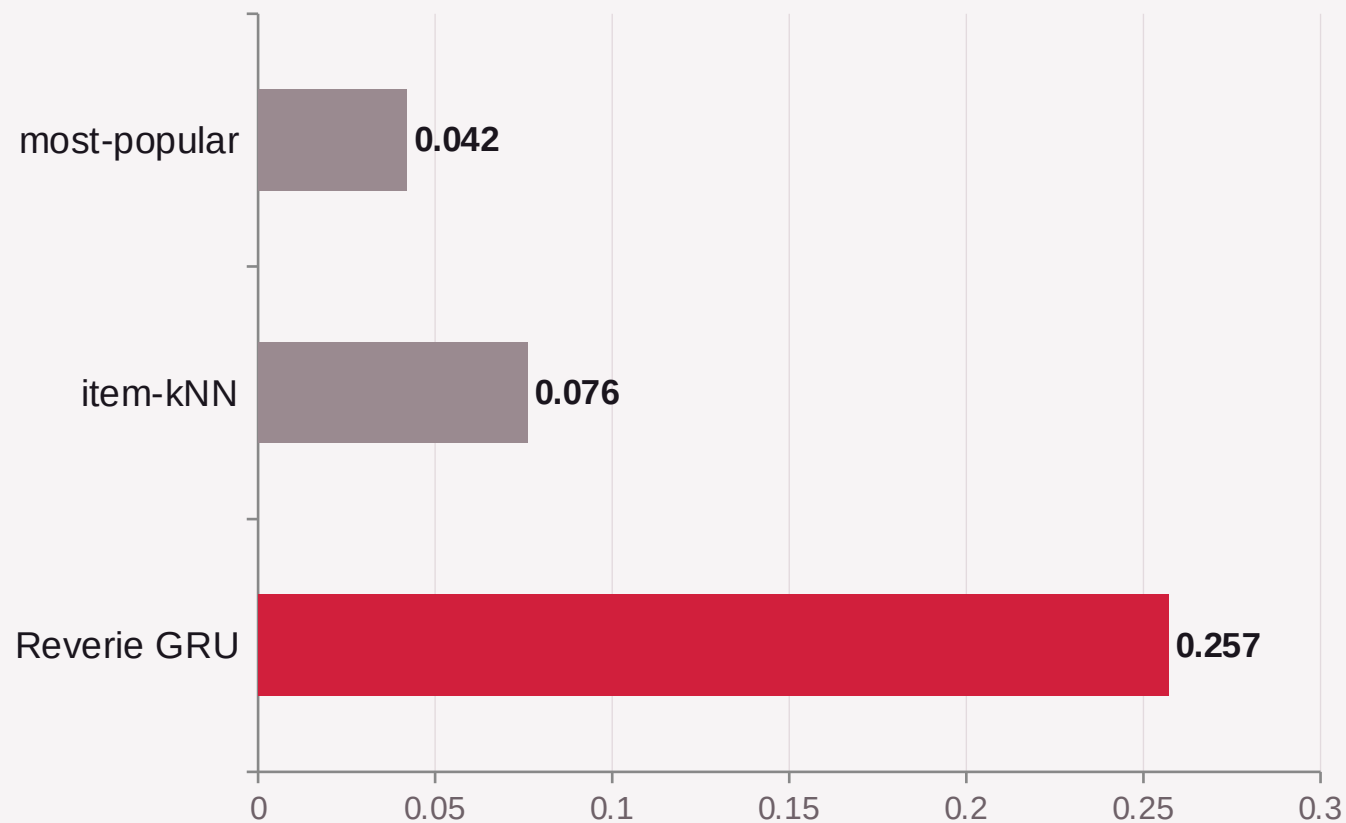
- It wins on MRR (0.135), the ranking-quality metric — and powers the live thumbs-down.



Rating earns its place

- The rating channel is what lets a “no” re-rank instantly in the demo. Alone it's noisy; with genre it's gold.

Far beyond the baselines



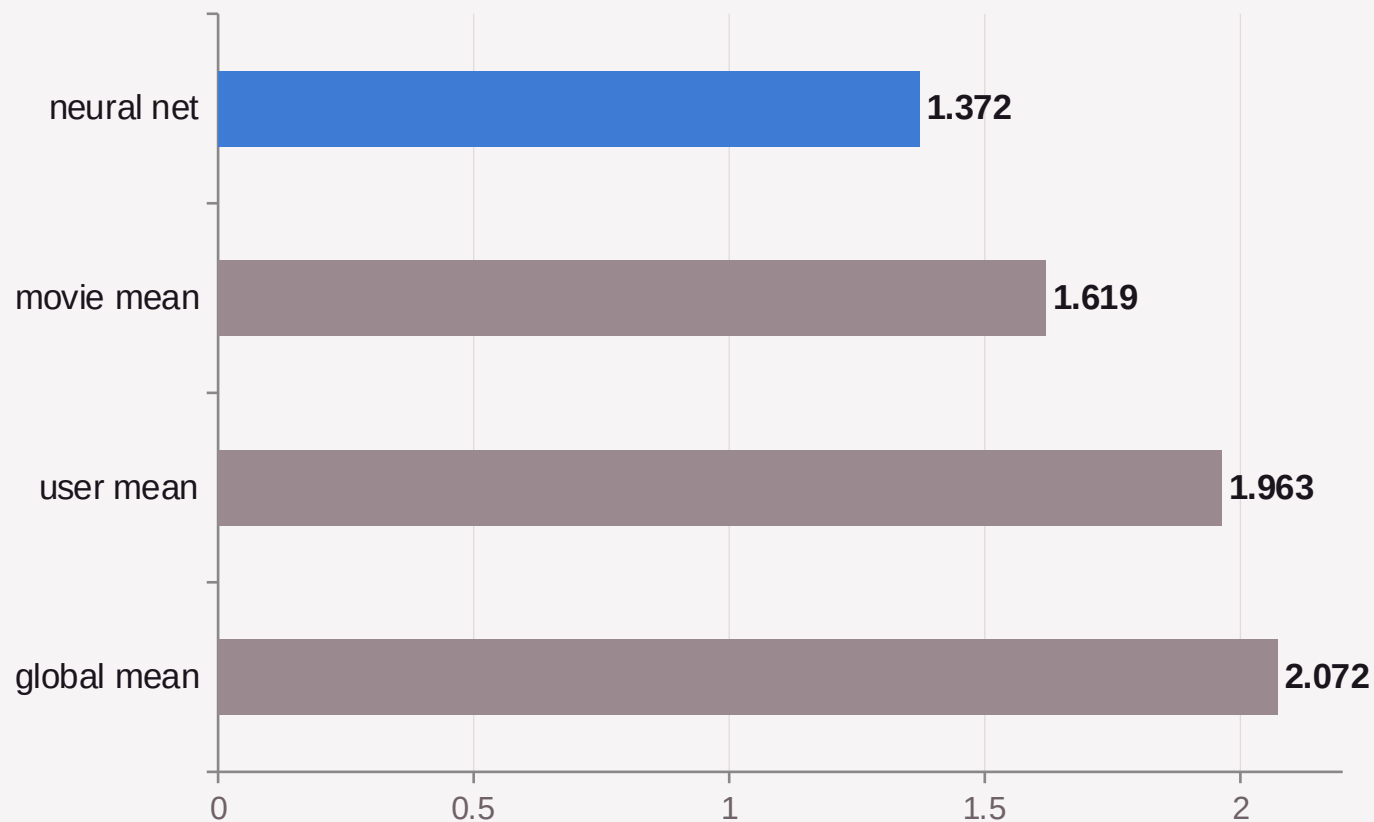
0.242

± 0.002

Frozen test set, 3 seeds.

Non-overlapping 95% CIs vs item-kNN — the strong baseline shallow RNNs often lose to.

Lower RMSE than every baseline



Lower is better

- The neural net (1.372) beats the movie-mean baseline for 94% of users — median per-user RMSE 1.32.



We're honest about it

- Cold start is still hard for brand-new viewers — and we say so rather than hide it.

Zero false alarms → precision 1.00

	predicted: not	predicted: love
actual: not	30 TN	0 FP = 0
actual: love	39 FN	4 TP



Conservative, but trustworthy

- Zero false alarms: every film flagged as a “love” actually was one.
- It misses some (39 false negatives) — but the short list it gives you is one you can trust.
- That trustworthy short list is exactly what a recommender needs.

03

The MVP

A real, working product

React talks to the model in under a second



A thumbs-down appends to your live history and re-ranks instantly — **with no retraining.**



Live MVP

See it in action.

Backup: if it doesn't play live, export demo.mov to .mp4 (H.264) and play the recording.

Lessons, known issues, next steps



Lessons learned

- Collaborative signal beats content — a single profile overfits.
- “Drama” tags half of all films; rank by nearest favorites instead.
- Honest baselines matter more than one flashy number.



Known issues

- Cold start: a brand-new viewer sits near their average.
- Embeddings lean to a Letterboxd cinephile cluster.
- Posters hotlink the Letterboxd CDN — a fragile dependency.



Next steps

- Per-user fine-tuning as a history grows.
- A mainstream-vs-deep-cuts dial.
- Group Blend beyond two people.



Thank you.

Questions?

Built on open data: MovieLens · samlearner & gsimonx37 (Letterboxd) · TMDB.

github.com/em-ech/reverie